

Worksheet: Effectiveness of liquid-liquid extraction

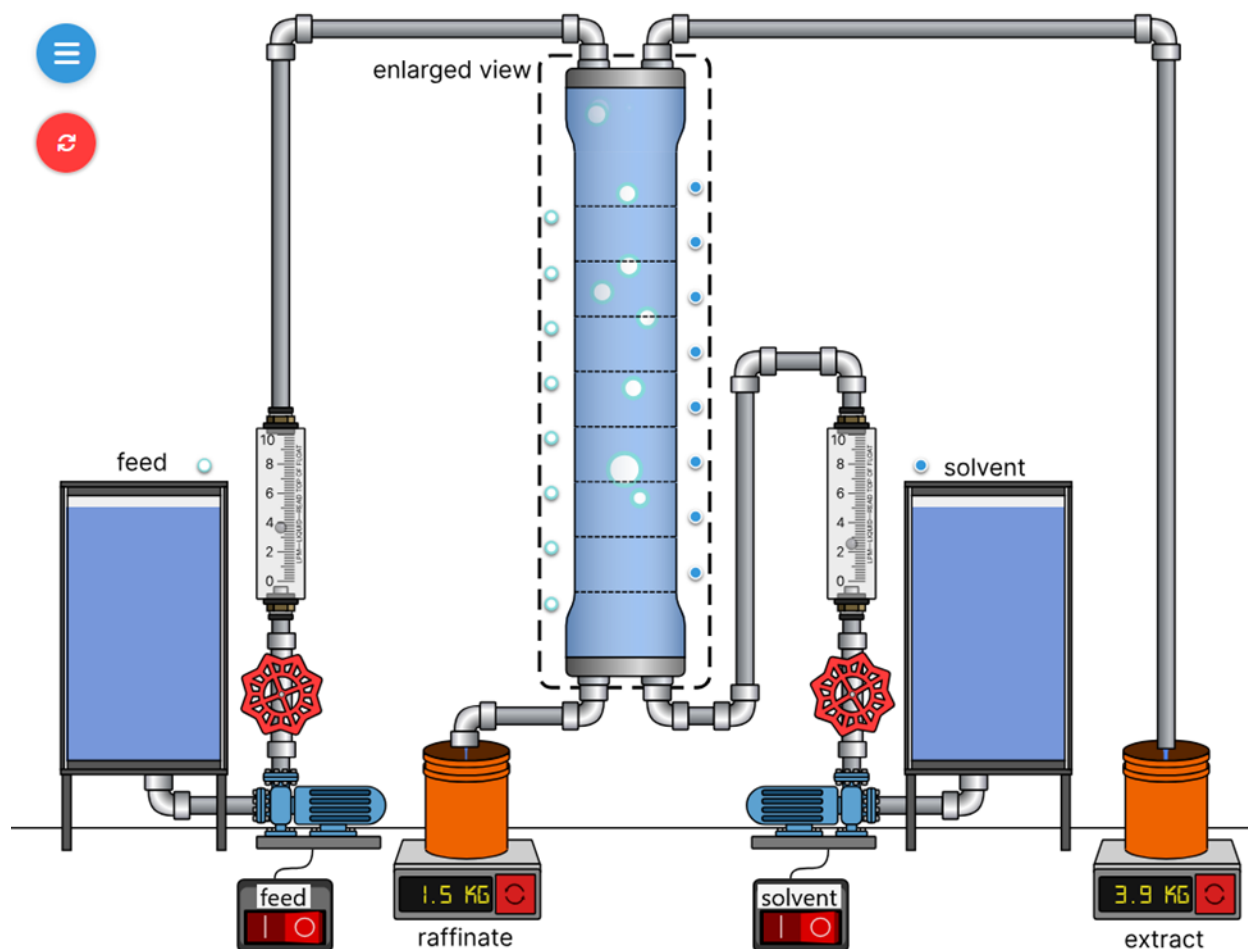
Name(s) _____

This experiment uses a steady-state liquid-liquid extraction process in which water as a solvent contacts an acetic acid/chloroform mixture. Because acetic acid is much more soluble in water, it is extracted from the chloroform organic phase into the aqueous phase.

Student learning objectives

1. Apply mass balances on a three-component system.
2. Determine the effects of the number of stages and stage efficiency on the separation efficiency.
3. Explain how changing the relative flow rates of feed and solvent affects the separation.
4. Generate a Hunter-Nash analysis on a ternary phase diagram.

Equipment



The number of stages can be varied between one and eight, and the stage efficiency can be varied from 50% to 100%. In each stage, the downward-flowing organic phase (acetic acid, chloroform, and a small concentration of water) contacts the upward-flowing aqueous solvent phase. The phases leaving each stage are in phase equilibrium. The water extracts acetic acid

solute from the organic feed. Thus, the extract that exits the top of the column contains water with acetic acid and a small amount of chloroform. Chloroform is enriched in the raffinate, which exits the bottom of the reactor, because acetic acid and water have been removed from it. Two buckets that sit on scales collect the flows out of the column.

Note that the feed and solvent tanks are large relative to the size of the column (thus the column is enlarged in the simulation view). Thus, the tank volumes do not change significantly during an experiment.

Notation

S = solvent mass flow rate

F = feed mass flow rate

E_n = mass flow rate of extract stream leaving stage n

R_n = mass flow rate of raffinate stream leaving stage n

$x_{F,i}$ = mass fraction of component i in feed stream

$y_{S,i}$ = mass fraction of component i in solvent stream

$x_{n,i}$ = mass fraction of component i in raffinate stream leaving stage n

$y_{n,i}$ = mass fraction of component i in extract stream leaving stage n

ρ_i = density of component i .

ρ_{mix} = density of mixture

K_D = distribution coefficient (or partition coefficient)

Questions to answer before starting the experiment

1. How will increasing the number of stages affect the separation?
2. How will decreasing stage efficiency affect the separation?
3. If the solvent flow rate decreases, how is the separation affected?

Experiment procedure:

Three distinct experiments are described below.

1. The mass percentages of the three components in the feed and their densities are:

Table 1: Feed composition and density

Component	Mass percent	Density (kg/L)
Chloroform	52.0	1.49
Acetic acid	43.0	1.04
Water	5.0	1.00
Feed		

The solvent only contains water.

Assume the density of a mixture is $\rho_{mix} = 1/\sum(x_i/\rho_i)$, where x_i is the mass fraction and ρ_i is the density of component i . Calculate fluid density in the feed tank and record it in Table 1.

2. When the experiments starts up, the column internals pop up is displayed; this allows the number of stages and the stage efficiency to be chosen. For each experiment below, these values will be different. To change values, select the reset button.

Experiment #1: Effect of number of stages

1. Select the number of stages as 2 and the stage efficiency as 100%. Turn on the solvent pump by clicking its I/O switch to fill the column with solvent. Only after the column fills, turn on the feed pump by clicking the I/O switch. Adjust the red feed and solvent valves by turning them. Select valve openings so that the feed flow rate is significantly higher than the solvent flow rate.
2. Record the volumetric flow rates for the feed and solvent and calculate the mass flow rates and record those values in Table 2.
3. Allow time for the system to reach steady state (when the mass percentages, measured by mousing over the white or blue dots, stop changing) and liquid to accumulate in the extract and raffinate tanks. Then determine the mass flow rates of the two exit streams by recording the mass change in the buckets while recording the time elapsed with your phone. Record values in Table 2.

Table 2: Volumetric and mass flow rates for 2 stages, 100% efficient

Tank	Volumetric flow rate (L/min)	Density (kg/L)	Elapsed time (s)	Mass change (kg)	Mass flow rate (kg/min)
Feed					$F =$
Solvent		1.00			$S =$
Extract					$E_1 =$
Raffinate					$R_2 =$

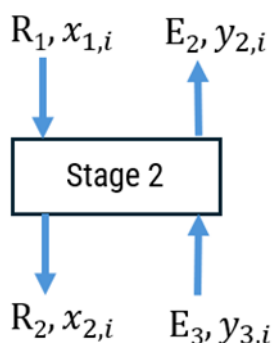
4. Do an overall mass balance on the system. Enter the mass flow rates of feed (F) and solvent (S) in Table 3 and add them together. Enter the mass flow rates of raffinate (R_2) and extract (E_1) out of the column in Table 3 and add them together. Calculate the difference and the percentage difference (relative to the total flow in) and enter those values in Table 3.

Table 3 Overall mass balances for 2 stages, 100% efficient

Mass flow rates (kg/min)							
Mass flow in			Mass flow out				
F	S	$F + S$	R_2	E_1	$R_2 + E_1$	Difference: in-out	% difference

Here are labels for mass flow rates and mass fractions entering and leaving stage 2.

R_n and E_n are mass flow rates; $x_{n,i}$ and $y_{n,i}$ are mass fractions.



5. The dots on the sides of the column measure the compositions leaving each stage. On the left are the organic phase compositions (raffinate, white dots) and on the right are the aqueous phase compositions (extract, blue dots). Likewise, the dots above the two tanks show the feed and solvent compositions. The top stage in the column is stage 1. Record the flow rate F and mass fraction of acetic acid into the column in Table 4. Record the flow rate of the extract leaving the tank (E_1) and the mass fraction of acetic acid in the extract

in Table 4. Calculate the fraction of the acetic acid in the feed that is extracted into the solvent and record in Table 4.

Table 4 Fraction of acetic acid removed from feed for 100% efficient stages

Number of stages	Feed			Extract			Fraction of acetic acid extracted
	F (kg/min)	$x_{F,A}$	$x_{F,A}F$	E_1 (kg/min)	$y_{1,A}$	$y_{1,A}E_1$	
2							
8							

- Select the number of stages as 8 and the stage efficiency as 100%. Turn on the solvent pump by clicking its I/O switch to fill the column with solvent. Only after the column fills, turn on the feed pump by clicking the I/O switch. Adjust the red feed and solvent valves by turning them. Select the same volumetric flow rates for the feed and solvent as used above.
- Allow time for the system to reach steady state and liquid to accumulate in the extract and raffinate tanks. Then determine the mass flow rate of the extract stream by recording the mass change in the bucket while recording the time elapsed with your phone. Record the values in Table 5 and calculate the extract mass flow rate (E_1). Record this value in Table 4. Add the mass fraction of acetic acid in the extract ($y_{1,A}$) and calculate the fraction of acetic acid extracted for 8 stages.

Table 5: Extract mass flow rates for 8 stages, 100% efficient

Tank	Elapsed time (s)	Mass change (kg)	Mass flow rate E_1 (kg/min)
Extract			

- For which system is the fraction extracted higher? Why?

9. Record the mass fractions for the raffinate and extract leaving each stage in Table 6. Calculate the distribution coefficient (K_D , also called the partition coefficient) for acetic acid in Table 6; this is the ratio of mass fraction of acetic acid in the aqueous phase to the mass fraction in the organic phase. For certain flow conditions, because of the composition locations on the phase envelope, oscillations occur.

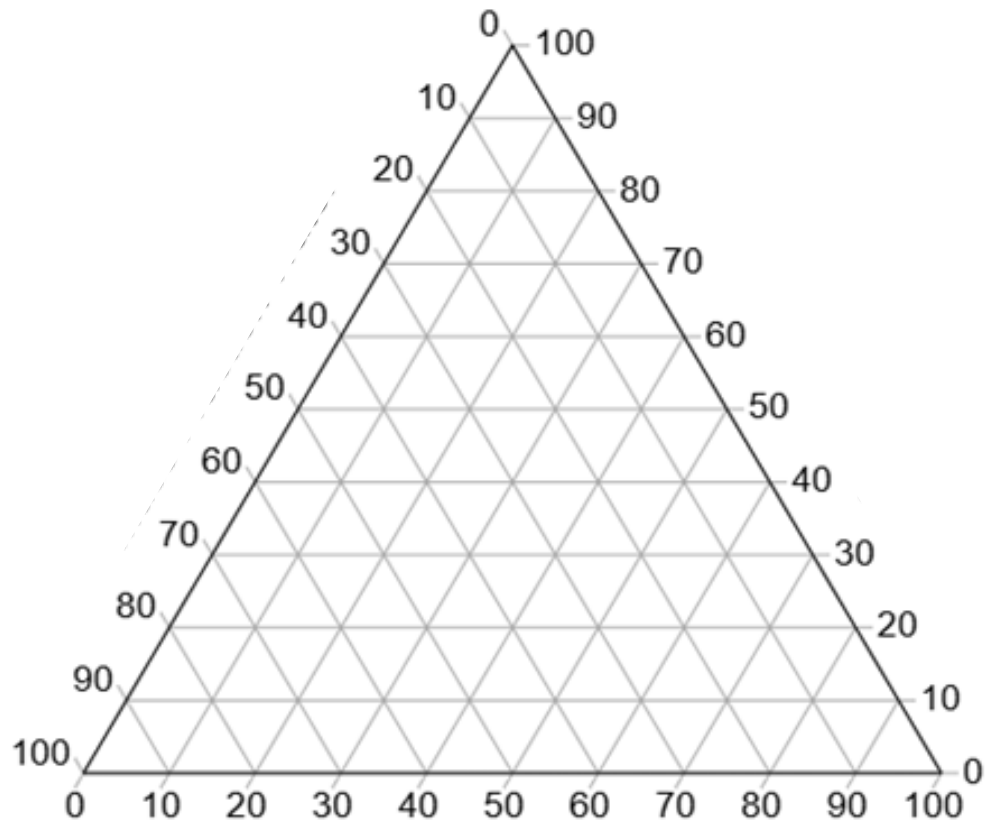
Table 6 Mass fractions leaving each stage and distribution coefficients for 8 stages, 100% efficient

Stage	Mass fractions raffinate			Mass fractions extract			K_D
	Acetic acid	Chloro-form	Water	Acetic acid	Chloro-form	Water	$\frac{y_{n,A}}{x_{n,A}}$
1							
2							
3							
4							
5							
6							
7							
8							

10. Are the distribution coefficients approximately constant? This is expected for dilute solutions. Note that when the acetic mass fractions are less than 1.0%, the K_D values are not accurate because the mass fractions are only measured to one digit after the decimal point.
11. Locate the compositions in Table 6 on the ternary phase diagram below and draw tie lines connecting compositions that are in equilibrium. Draw the phase envelope; points connected by tie lines are on the phase envelope.

12. Use the results for 8 stages to make a Hunter-Nash plot on the ternary phase diagram. In addition to the above, this involves locating the feed and solvent concentrations on the ternary phase diagram, identifying the mixing point, and stepping off stages. This simulation explains a Hunter-Nash diagram:

<https://learncheme.com/simulations/separations/hunter-nash-method-1le/> .



Experiment #2: Effect of stage efficiency

1. Select the number of stages as 8 and the stage efficiency as 60%. Turn on the solvent pump by clicking its I/O switch to fill the column with solvent. Only after the column fills, turn on the feed pump by clicking the I/O switch. Adjust the red feed and solvent valves by turning them. Select the same volumetric flow rates for the feed and solvent as used for Experiment #1 for 8 stages.
2. Allow time for the system to reach steady state and liquid to accumulate in the extract and raffinate tanks. Then determine the mass flow rate of the extract stream by recording the mass change in the bucket while recording the time elapsed with your phone. Record the values in Table 7 and calculate the extract mass flow rate (E_1). Record this value in Table 8. Record the mass fraction of acetic acid in the extract ($y_{1,A}$) in Table 8, add the feed flow rate and mass fraction of acetic acid in the feed, and calculate the fraction of acetic acid extracted for 8 stages with 60% efficiency.

Table 7: Extract mass flow rates for 8 stages, 60% efficient

Tank	Elapsed time (s)	Mass change (kg)	Mass flow rate E_1 (kg/min)
Extract			

Table 8 Fraction of acetic acid removed from feed for 8 stages, 60% efficient

Feed			Extract			Fraction of acetic acid extracted
F (kg/min)	$x_{F,A}$	$x_{F,A} F$	E_1 (kg/min)	$y_{1,A}$	$y_{1,A} E_1$	$\frac{y_{1,A} E_1}{x_{F,A} F}$

3. Compare the fraction extracted for 8 stages for 100% and 60% stage efficiency. Discuss the difference.

Experiment #3: Effect of feed to solvent ratio

1. Select the number of stages as 5 and the stage efficiency as 100%. Turn on the solvent pump by clicking its I/O switch to fill the column with solvent. Only after the column fills, turn on the feed pump by clicking its I/O switch. Adjust the red feed and solvent valves by turning them. Note: only the feed rate in this experiment.
2. Select a solvent flow rate of about 1.0 L/min. Record this value in Table 9 and calculate its mass flow rate. Use this flow rate for all the measurements in this experiment.
3. Select a feed flow rate about 3 L/min (low row in Table 9). Record the volumetric flow rate for the feed and calculate the mass flow rate and record those values in Table 9.
4. Allow time for the system to reach steady state and liquid to accumulate in the extract and raffinate tanks. Then determine the mass flow rates of the extract stream by recording the mass change in the buckets while recording the time elapsed with your phone. Record values in Table 9.

Table 9: Mass flow rates for 5 stages, 100% efficient

Feed flow rate	Tank	Volumetric flow rate (L/min)	Density (kg/L)	Elapsed time (s)	Mass change (kg)	Mass flow rate (kg/min)
low	Feed					$F =$
	Extract					$E_1 =$
medium	Feed					$F =$
	Extract					$E_1 =$
high	Feed					$F =$
	Extract					$E_1 =$
	Solvent		1.00			$S =$

5. Record the feed and extract mass flow rates and the F/S ratio in Table 10.
6. Record the mass fractions of acetic acid in the feed and in the extract in Table 10.
7. Without changing the solvent flow rate, change the feed flow rate to about 6 L/min (middle row in Table 9) and repeat steps 3 and 4. Record the feed and extract mass flow rates and the F/S ratio in Table 10. Record the mass fraction of acetic acid in the extract in Table 10.
8. Without changing the solvent flow rate, change the feed flow rate to about 9 L/min (high row in Table 9) and repeat steps 3 and 4. Record the feed and extract mass flow rates and the F/S ratio in Table 10. Record the mass fraction of acetic acid in the extract in Table 10.

Table 10 Fraction of acetic acid removed from feed for different feed flow rates

Feed flow rate	Feed			Extract			Fraction of acetic acid extracted	F/S
	F (kg/min)	$x_{F,A}$	$x_{F,A}F$	E_1 (kg/min)	$y_{1,A}$	$y_{1,A}E_1$	$\frac{y_{1,A}E_1}{x_{F,A}F}$	
low								
medium								
high								

9. Discuss how the F/S ratio affects the fraction of acetic acid extracted.

Questions to answer

1. Where might these measurements have errors?
2. What safety precautions would you take to conduct this experiment in the laboratory?
3. At what conditions (number of stages, flow rates) would you recommend running this liquid-liquid extraction to extract the largest fraction of acetic acid ? Why?
4. What matters more in this experiment: number of stages or feed-to-solvent ratio?